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Unit – 4 ⇝ Advanced Matrix Algebra (Lecture Note)

Introduction

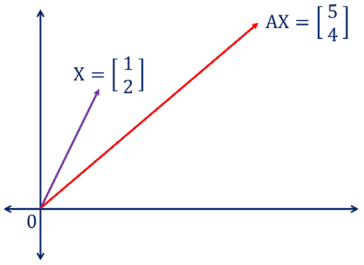
→ In this unit, we explore some advanced concepts of matrix algebra that help us to understand the behavior of matrices more deeply.

→ The concepts of **eigenvalues** and **eigenvectors** describe special directions that remain **unchanged** under a linear transformation, except for a change in scale.

→ **Case - 1**

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, then $AX = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$

i.e., the resultant vector has been scaled and rotated.

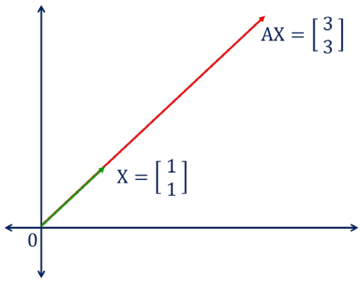


- Here, the direction of AX is different from the direction of X.
- Therefore, the matrix transformation changes both the **length** and the **direction** of the vector. Because, X is **not an eigenvector** of A.

→ **Case - 2**

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then $AX = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

i.e., the resultant vector has been scaled but not rotated.



- Here, the transformed vector AX lies on the same line as the original vector X.
- The matrix transformation only changes the **length** of the vector by a factor of 3, but its direction remains **unchanged**.
- Therefore, X is an **eigenvector** of A, and the corresponding **eigenvalue** is 3.

→ **Cayley-Hamilton Theorem**, which establishes a fundamental link between a matrix and its characteristic equation.

→ we will learn about **quadratic forms** and **canonical forms**, which provide simpler representations of matrices and help in solving complex problems efficiently.

→ Eigenvalues and eigenvectors are widely used in Artificial Intelligence, Machine Learning, data analysis, image processing, and pattern recognition.

Method 1 ⇝ Eigenvalues and Eigenvectors

Eigenvalue & Eigenvector

→ Let A be a square matrix.

→ If there exists a non-zero vector X and scalar λ, such that $\mathbf{AX} = \boldsymbol{\lambda}\mathbf{X}$, then $\boldsymbol{\lambda}$ is known as an eigenvalue of A and X is known as an eigenvector of A corresponding to eigenvalue λ.

→ The set of all eigenvalues of the matrix A is known as **Spectrum** of A.

→ Eigenvalue is also known as a characteristic root, latent value or proper roots.

→ Eigenvector is also known as a characteristic vector, latent vector or proper vector corresponding to the eigenvalue λ of the matrix.

Characteristic Equation

→ Let A be a square matrix of order n and I be an identity matrix of order n, then $|\mathbf{A} - \boldsymbol{\lambda}\mathbf{I}| = \mathbf{0}$ is known as characteristic equation, where λ is a scalar.

→ The characteristic equation of matrix A of order 2×2 is

$$\boldsymbol{\lambda^2 - S_1\lambda + S_2 = 0}$$

Where $S_1 = \text{tr}(A) =$ Sum of diagonal entries

$S_2 = |A| =$ Determinant of A

→ The characteristic equation of matrix A of order 3×3 is

$$\boldsymbol{\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0}$$

Where $S_1 = \text{tr}(A) =$ Sum of diagonal entries

$S_2 =$ Sum of minors of principal diagonal elements $= M_{11} + M_{22} + M_{33}$

$S_3 = |A| =$ Determinant of A

→ The **roots** of characteristic equation are known as **eigenvalues** of matrix A.

→ **Example**: Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$.

Solution:

The characteristic equation of the matrix A is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ ... (1)

where,

$\mathbf{S_1} = \text{tr}(A) = 2 + 2 + 0 = 4$

$\mathbf{S_2}$ = Sum of minors of principal diagonal elements

$$= \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix}$$

$$= 1 + 0 + 4 = 5$$

$\mathbf{S_3}$ = | A |

$$= \begin{vmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{vmatrix} = 2(0 + 1) - 0 + 0 = 2$$

→ Put the values of S_1, S_2 and S_3 in equation (1)

$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$; Which is required characteristic equation.

Eigen Space

→ The **solution** space of the system $\mathbf{(A - \lambda I)X = O}$ is known as an **Eigen Space** of matrix A corresponding to eigenvalue λ.

i.e., $E_\lambda = \{ X \mid (A - \lambda I)X = O \}$

Procedure to find the eigenvalues and eigenvectors for a matrix A of order $\mathbf{3 \times 3}$

(1) Write the characteristic equation $\mathbf{\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0}$ and find S_1, S_2 and S_3.

(2) Find the **roots** of the characteristic equation. These are the Eigenvalues of A, say λ_1, λ_2 and λ_3.

(3) Solve the homogeneous system $(A - \lambda I)X = O$ for each eigenvalue by using **Gauss Elimination** Method.

(4) **Eigenvector** is the solution **X** of the above system.

→ **The procedure remains same for any matrix of order n.**

Properties of Eigenvalue and Eigenvector

(1) $\mathbf{Trace(A)}$ = Sum of all the eigenvalues of matrix A.

(2) $\mathbf{|A|}$ = Product of all the eigenvalues of matrix A.

(3) If one of the eigenvalue of matrix A is zero, then | A | = 0, hence A^{-1} does not exist.

(4) The eigenvalues of triangular matrix are the elements on its **principal diagonal.**

(5) The square matrix A and A^T have the same eigenvalues but eigenvectors need not be the same.

(6) If λ is an eigenvalue of a non-singular matrix A and X is an eigenvector corresponding to λ, then

Matrix	Eigenvalue
A^T	λ
A^k	λ^k ; $k \in \mathbb{Z} - \{0\}$
A^{-1}	$\frac{1}{\lambda}$
kA	$k\lambda$; $k \in \mathbb{R}$
$A \pm kI$	$\lambda \pm k$; $k \in \mathbb{R}$
$A^m \pm kI$	$\lambda^m \pm k$; $k \in \mathbb{R}$

Examples of Method-1: Eigenvalues and Eigenvectors

C	1	Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$. **Answer:** $\boldsymbol{\lambda = 2,\ 2,\ 2,\quad X = \left[-\frac{1}{3}\quad -\frac{1}{3}\quad 1\right]^T}$
C	2	If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then find the eigenvalues and eigenvectors corresponding to each eigenvalue of A. Also, write the eigen space for each eigenvalue. **Answer:** $\boldsymbol{\lambda = -1 \rightsquigarrow X = [-1\quad 1]^T,\quad \lambda = 1 \rightsquigarrow X = [1\quad 1]^T}$ $\boldsymbol{E_{\lambda=-1} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\},\quad E_{\lambda=1} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\}}$
C	3	Find the eigenvalues of $A = \begin{bmatrix} -5 & 4 & 34 \\ 0 & 0 & 4 \\ 0 & 0 & 4 \end{bmatrix}$. Is it invertible? **Answer:** $\boldsymbol{-5,\ 0,\ 4}$ & **No**

C	4	Let $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$, then find eigenvalues of A, A^{25}, 3A, A^{-1}, A^T, $A + 2I$ & $A^3 - 5I$. **Answer:** $\mathbf{A \rightsquigarrow -1,\ 3}$; $\mathbf{A^{25} \rightsquigarrow -1,\ 3^{25}}$; $\mathbf{3A \rightsquigarrow -3,\ 9}$ $\mathbf{A^{-1} \rightsquigarrow -1,\ \frac{1}{3}}$; $\mathbf{A^T \rightsquigarrow -1,\ 3}$; $\mathbf{A + 2I \rightsquigarrow 1,\ 5}$ $\mathbf{A^3 - 5I \rightsquigarrow -6,\ 22}$

Method 2 ⇝ Cayley – Hamilton Theorem

Cayley – Hamilton Theorem

→ Every square matrix **satisfies** its own **characteristic equation.**

→ **Example**: For matrix $A = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}$, characteristic equation is $\lambda^2 - 6\lambda - 7 = 0$.

Substitute $\lambda = A$, to check whether A satisfies characteristic equation or not.

$$A^2 - 6A - 7I = \begin{bmatrix} 13 & 18 \\ 24 & 37 \end{bmatrix} - 6\begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix} - 7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 18 \\ 24 & 37 \end{bmatrix} - \begin{bmatrix} 6 & 18 \\ 24 & 30 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

→ Using the Cayley-Hamilton theorem, we can find any integer power of A.

i.e., A^n; $n \in \mathbb{Z}$.

Procedure to verify Cayley-Hamilton Theorem for 3×3 matrix

(1) Find the characteristic equation $\boldsymbol{\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0}$ of given matrix A.

(2) Put $\boldsymbol{\lambda = A}$ in the characteristic equation. i.e. $\boldsymbol{A^3 - S_1A^2 + S_2A - S_3I = 0}$ ...(1).

(3) Find A^3, A^2 using given matrix A.

(4) Put the values of A^3, A^2, A, I in equation (1) and verify it.

→ **The procedure remains same for any matrix of order n.**

→ Note:

- To find A^{-1} using Cayley – Hamilton theorem (whenever A^{-1} exists)
  - Multiply characteristic equation both sides by A^{-1}
  - Substitute the values of $A^{n-1}, A^{n-2}, \ldots, A$ and I.
- By the similar way we can also find A^{-2}, A^2, A^4 etc ...

Examples of Method-2: Cayley – Hamilton Theorem

C	1	Verify Cayley – Hamilton theorem and use it to find the inverse of $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. **Answer:** $\frac{1}{4}\begin{bmatrix} \mathbf{3} & \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{3} & \mathbf{1} \\ \mathbf{-1} & \mathbf{1} & \mathbf{3} \end{bmatrix}$
C	2	Using Cayley-Hamilton theorem for matrix $A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -1 & 4 \\ -2 & 1 & 5 \end{bmatrix}$, find matrix represented by $A^8 - 5A^7 - A^6 + 37A^5 + A^4 - 5A^3 - 3A^2 + 41A + 3I$. **Answer:** $\begin{bmatrix} \mathbf{13} & \mathbf{8} & \mathbf{-40} \\ \mathbf{16} & \mathbf{-11} & \mathbf{-16} \\ \mathbf{16} & \mathbf{8} & \mathbf{-27} \end{bmatrix}$

Method 3 ⇝ Algebraic Multiplicity and Geometric Multiplicity

Algebraic Multiplicity (AM)

→ The algebraic multiplicity of an eigenvalue is the **number of times** it appears as a root of the characteristic polynomial of a matrix.

Geometric Multiplicity (GM)

→ Geometric multiplicity of an eigenvalue is the **dimension of the eigenspace (free variables)** associated with an eigenvalue.

or

Geometric multiplicity of an eigenvalue λ of the matrix A is **nullity of $A - \lambda I$.** i.e. $\eta(A - \lambda I)$.

→ **Remark:** $1 \leq GM \leq AM$.

i.e. If AM of $\lambda = 1$, then GM of $\lambda = 1$.

→ **Example:**

Find the geometric multiplicity of the eigenvalues of a matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

Solution:

The characteristic polynomial of a matrix A is,

$\boldsymbol{\lambda^2(\lambda - 3) = 0}$

$\Rightarrow \boldsymbol{\lambda_1 = \lambda_2 = 0}$ and $\boldsymbol{\lambda_3 = 3}$.

So, AM of eigenvalue 0 is **2 (appears twice)** and AM of eigenvalue 3 is **1 (appears once)**.

Here, $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 = 3$.

After solving the system $(A - \lambda I)X = O$ for each eigenvalue,

The eigenspace of $\lambda_1 = 0$ is $\mathbf{E_{\lambda_1=0}} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \cdot \mathbf{s} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \cdot \mathbf{t} \;\middle|\; \mathbf{s},\ \mathbf{t} \in \mathbb{R} \right\}$

The eigenspace of $\lambda_3 = 3$ is $\mathbf{E_{\lambda_3=3}} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \mathbf{t} \;\middle|\; \mathbf{t} \in \mathbb{R} \right\}$.

Here, there are 2 LI vectors to represent $E_{\lambda_1=0}$. So, Dimension of $\mathbf{E_{\lambda_1=0}} = \mathbf{2}$.

Here, there are 1 LI vector to represent $E_{\lambda_3=3}$. So, Dimension of $\mathbf{E_{\lambda_3=3}} = \mathbf{1}$.

So, the GM of eigenvalue 0 is **2** and eigenvalue 3 is **1**.

Alternatively,

- $A - \lambda_1 I = A - 0I = A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$.

  So, $\eta(A - 0I) = \text{No. of column} - \rho(A - 0I) = 3 - 1 = 2$

  So, the GM of eigenvalue 0 is **2**.

- $A - \lambda_3 I = A - 3I = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$

  So, $\eta(A - 3I) = \text{No. of column} - \rho(A - 3I) = 3 - 2 = 1$

  So, the GM of eigenvalue 3 is **1**.

Examples of Method-3: Algebraic Multiplicity and Geometric Multiplicity

C	1	Find AM and GM for each of the eigenvalue of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$. **Answer: For $\lambda = 1$: AM = 1 and GM = 1** **For $\lambda = 2$: AM = 2 and GM = 1**
C	2	Find $\eta(A - \lambda I)$ for each λ of $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$. **Answer: $\eta(A - \lambda I) = 1$ for each λ.**
C	3	Let $A_{5\times5}$ be a matrix with $\lvert A \rvert = 720$, $\text{tr}(A) = 15$ and the eigenvalues of the matrix A are 2, 3, 5, α and β then find the possible values of $\eta(A - 5I)$. **Answer: $\eta(A - 5I) = 1$**

Method 4 ⇝ Diagonalization of Matrices

Matrix Diagonalization

→ A square matrix A is **diagonalizable** if it is similar to diagonal matrix D.

or

A square matrix A is **diagonalizable** if there exist an invertible matrix P and diagonal matrix D such that $D = P^{-1}AP$.

→ Note:

- The invertible matrix P is known as **diagonalizing matrix/modal matrix**.
- A diagonal matrix D is known as a **spectral matrix.**
- A square matrix A is **orthogonally diagonalizable** if there exist an **orthogonal matrix P′** and diagonal matrix D such that $D = P'^{-1}AP'$.

Properties

- If all the **eigenvalues** of a square matrix A are **distinct**, then a matrix A is always diagonalizable.
- A square matrix A of order n is diagonalizable if and only if it possesses **n linearly independent eigenvectors**.
- A square matrix A is diagonalizable if and only if the **AM** is **equal** to the **GM** for each eigenvalue.
- Every **real symmetric** matrix is always diagonalizable as well as orthogonally diagonalizable.
- For **real skew-symmetric** matrix A, $\mathbf{A^{2n}}$ is always diagonalizable.
- An **Idempotent** and **Involutory** matrices are always diagonalizable.

Procedure to find a Diagonalizing Matrix / Modal Matrix (P)

Let A be any square matrix of order n.

(1) Find the **eigenvalues** ($\lambda_1, \lambda_2, \ldots, \lambda_n$) of matrix A.

(2) Find **eigenvectors** corresponding to each eigenvalue of the matrix A.

(3) Construct the **diagonalizing matrix (P)** by placing all the eigenvectors into the **columns** of the matrix.

→ Note:

- If $\mathbf{P^{-1}}$ **exist**, then the diagonal matrix $\mathbf{D = diag[\lambda_1, \lambda_2, \ldots, \lambda_n]}$.

Procedure to find a Spectral Matrix / Diagonal Matrix (D)

Let A be any square matrix of order n.

(1) Find the **eigenvalues** ($\lambda_1, \lambda_2, \dots, \lambda_n$) of matrix A.

(2) Find **AM** & **GM** for each eigenvalue of the matrix A.

(3) If **AM** & **GM** for each eigenvalue is equal, then diagonal matrix

$\mathbf{D = diag[\lambda_1, \lambda_2, \dots, \lambda_n]}$

Examples of Method-4: Diagonalization of Matrices

C	1	Find diagonalizing matrix and diagonal matrix for a given matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. **Answer:** $\mathbf{P} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
C	2	Show that the matrix $A = \begin{bmatrix} 1 & -2 & 0 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ is not diagonalizable.

Method 5 ⇝ Quadratic Forms and Canonical Forms

Unit Vector

→ Let $x = (x_1, x_2, \dots, x_n)$ be any non-zero vector of $\mathbb{R}^n$. The **length** of vector x is defined as $\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$.

→ Any non-zero vector $x = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ can be made a **unit vector** by dividing its length. i.e.

$$x' = \textbf{Unit Vector} = \frac{(x_1, x_2, \dots, x_n)}{\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}}$$

Quadratic Forms

→ A **quadratic form** is a function $Q : \mathbb{R}^n \to \mathbb{R}$ defined as $\mathbf{Q(x) = x^T Ax}$, where x is a vector from $\mathbb{R}^n$ and A is an $n \times n$ symmetric matrix.

Here, the matrix A is called the **matrix of quadratic form.**

Matrix of Quadratic Form

→ The quadratic form corresponding to a symmetric matrix A in n variables $x_1, x_2, \dots, x_n$ can be written as

$$\mathbf{Q(x) = x^T Ax} = \sum_{i=1}^{n}\sum_{j=1}^{n} \mathbf{a_{ij} x_i x_j} \quad \dots (1)$$

→ Where $a_{ij} = a_{ji}$ for each i and j and

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{12} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{bmatrix}$$

→ From equation (1),

- Coefficient of $x_i x_j = a_{ij} + a_{ji} = 2a_{ij} = 2a_{ji}$ $\left(\because \mathbf{a_{ij} = a_{ji}\ ; i \neq j} \right)$
- Coefficient of $x_i^2 = a_{ii}$

→ **Remarks:**

(1) The general Quadratic form for 2 variables is given as,

- $Q(x_1, x_2) = a_{11}x_1^2 + a_{22}x_2^2 + 2a_{12}x_1x_2$.

(2) The general Quadratic form for 3 variables is given as,

- $Q(x_1, x_2, x_3) = a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + 2a_{12}x_1x_2 + 2a_{13}x_1x_3 + 2a_{23}x_2x_3$.

→ **Example:** Find a quadratic form for a matrix $A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$.

Solution:

For 2×2 matrix, Quadratic Form is $a_{11}\, x_1^2 + 2a_{12}\, x_1x_2 + a_{22}\, x_2^2$

- Coefficient of $x_1^2 = a_{11} = 3$
- Coefficient of $x_2^2 = a_{22} = 7$
- Coefficient of $x_1x_2 = 2a_{12} = -4$

Now, $Q(x) = a_{11}\, x_1^2 + 2a_{12}\, x_1x_2 + a_{22}\, x_2^2 = 3x_1^2 - 4x_1x_2 + 7x_2^2$

Change of Variable in a Quadratic Form

- We have a quadratic form $\mathbf{Q(x) = x^TAx}$, where x is a vector from $\mathbb{R}^n$ and A is an $n \times n$ symmetric matrix.
- Change of variable is an equation of the form $\mathbf{x = P'y}$ or $\mathbf{y = P'^{-1}x}$ … (2)

  Where, a matrix P′ is obtained by placing all the **unit eigenvectors** into the **columns** of the matrix and y is a new variable vector in $\mathbb{R}^n$
- Here, matrix P′ becomes orthogonal. so eq^n (2) is called an **orthogonal change of variable**.
- Since P′ is orthogonal matrix, $\mathbf{P'^{-1} = P'^{T}}$.
- Now put the value of $\mathbf{x = P'y}$ in a quadratic form as follows.

$$Q(x) = x^TAx$$
$$= (P'y)^TA(P'y)$$
$$= y^T P'^T A P'y$$
$$= y^T(P'^TAP')y$$
$$= y^T(P'^{-1}AP)y \qquad (\because P'^{-1} = P'^T)$$
$$= y^TDy \ \ldots(3) \qquad (\because P'^{-1}AP' = D = P'^TAP')$$

→ **Procedure to find Orthogonal Change of Variable**

(1) Construct the **symmetric matrix A** from given quadratic form.

(2) Find **eigenvectors** corresponding to each eigenvalue of the matrix A.

(3) Convert all eigenvectors into **unit eigenvectors** by dividing its length.

(4) Construct the **orthogonal matrix (P′)** by placing all the unit eigenvectors into the **columns** of the matrix.

(5) The equation $\mathbf{x = P'y}$ is an orthogonal change of variable.

Canonical Form or Normal Form

→ Let $Q(x) = x^T Ax$ be a quadratic form. By changing the variable $x = P'y$, quadratic form transforms to

$$\mathbf{Q(y) = y^T Dy = \sum_{i=1}^{r} \lambda_i y_i^2} \quad (\because \text{Equation (3)})$$

in new variables.

Where $\boldsymbol{\lambda_i}$ **are eigenvalues** of the matrix A.

→ This is the new quadratic form containing only the squares of y_i is called the **canonical form** or **normal form** of the given quadratic form.

→ **Procedure to find Canonical Form of a Quadratic Form**

(1) Construct the **symmetric matrix A** of order n from given quadratic form.

(2) Find the **eigenvalues** $(\boldsymbol{\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n})$ of the matrix A.

(3) Use eigenvalues to find **Canonical Form** of the given quadratic form.

i.e. $Q = y^T Dy = \boldsymbol{\lambda_1}\, y_1^2 + \boldsymbol{\lambda_2}\, y_2^2 + \boldsymbol{\lambda_3}\, y_3^2 + \cdots + \boldsymbol{\lambda_n}\, y_n^2$

Rank of Quadratic Form

→ The **rank of the matrix A** is called the rank of the quadratic form and it is denoted by **r**.

- In other words, the number of non-zero eigenvalues of A is equal to the rank of the quadratic form of A.
- If A is an $n \times n$ symmetric matrix and $\boldsymbol{\rho(A) < n}$ $(\text{i.e.} \mid A \mid = 0)$,then the quadratic form is **singular**, otherwise it is **non-singular**.

Index of Quadratic Form

→ The number of **positive eigenvalues** of the matrix A is called the **index** of the quadratic form and it is denoted by **p**.

Signature of Quadratic Form

→ The **difference** between the numbers of **positive and negative eigenvalues** is called the **signature** of the quadratic form and it is denoted by **s.**

i.e. Signature $s = p - (r - p) = \mathbf{2p - r}$

Where, No. of Negative Eigenvalues $= r - p$

Value Class or Nature of Quadratic Form

→ Let $Q(x) = x^T Ax$ be the quadratic form in n variables $x_1, x_2, \dots, x_n$. Let **r** be a rank of A, $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of A and **p** be an index of Q(x).

→ Following are criteria for the definiteness of the value class of Q.

Sr. No.	Value Class	Q(x)
1.	Positive Definite	$Q(x) > 0, \ \forall\, x \neq 0$
2.	Positive Semidefinite	$Q(x) \geq 0, \ \forall\, x$
3.	Negative Definite	$Q(x) < 0, \ \forall\, x \neq 0$
4.	Negative Semidefinite	$Q(x) \leq 0, \ \forall\, x$
5.	Indefinite	$Q(x_1) < 0$ & $Q(x_2) > 0$ for some $x_1 \neq x_2$

→ **Criterial for the Value Class of a Quadratic Form in terms of the nature of Eigenvalues**

Sr. No	Value Class	Nature of Eigenvalues	Criteria
1.	Positive Definite	All positive	$r = p = n$
2.	Positive Semidefinite	Positive with at least one zero	$r = p, r < n$
3.	Negative Definite	All negative	$r = n, p = 0$
4.	Negative Semidefinite	Negative with at least one zero	$p = 0, r < n$
5.	Indefinite	Positive as well as Negative	$0 < p < r \leq n$

n is order of matrix

→ **Criteria for the Value Class of a Quadratic Form in terms of Leading Principal Minors**

- For the matrix $A = \begin{bmatrix} \mathbf{a_{11}} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$

- The leading principal minors of matrix A are those determinants starting with a_{11} of orders 1, 2, … n, i.e.

$$\Delta_1 = |\, \mathbf{a_{11}} \,|\,; \qquad \Delta_2 = \begin{vmatrix} \mathbf{a_{11}} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}; \qquad \Delta_3 = \begin{vmatrix} \mathbf{a_{11}} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots$$

$$\Delta_n = |\, A \,| = \begin{vmatrix} \mathbf{a_{11}} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

Sr. No	Value Class	Nature of Leading Principal Minors
1.	Positive Definite	$\Delta_k > 0$ for all $k = 1,2, \dots, n$.
2.	Negative Definite	$\Delta_k < 0$ for all odd k and $\Delta_k > 0$ for all even k.

→ **Note:**

- This method is applicable only to **positive definite** and **negative definite** cases.
- If the leading principal minors follow any **other sign pattern**, this method is **not applicable**.

Examples of Method-5: Quadratic Forms and Canonical Forms

C	1	Write down the Quadratic forms corresponding to the following matrices. (i) $A = \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & -2 \\ 5 & -2 & 4 \end{bmatrix}$, (ii) $B = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$ **Answer**: **(i)** $\mathbf{2x_1^2 + 3x_2^2 + 4x_3^2 + 2x_1x_2 + 10x_1x_3 - 4x_2x_3}$ **(ii)** $\mathbf{0x_1^2 + 2x_2^2 + 4x_3^2 + 6x_4^2 + 2x_1x_2 + 4x_1x_3 + 6x_1x_4 + 6x_2x_3 + 8x_2x_4 + 10x_3x_4}$
C	2	Express the quadratic form $Q(x) = 2x_1^2 + 3x_2^2 - 5x_3^2 - 2x_1x_2 + 6x_1x_3 - 10x_2x_3$ in matrix notation. **Answer**: $\mathbf{x^TAx} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ -1 & 3 & -5 \\ 3 & -5 & -5 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$
C	3	Find an orthogonal change of variable that eliminates the cross-product terms in the quadratic form $Q(x) = x_1^2 - x_3^2 - 4x_1x_2 + 4x_2x_3$. **Answer**: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$,
C	4	Reduce the quadratic form $Q(x) = 3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_2x_3 + 2x_3x_1$ to canonical form by orthogonal transformation. **Answer**: **Canonical form** $\mathbf{Q(y) = 2y_1^2 + 3y_2^2 + 6y_3^2}$

C	5	Determine the rank, index, signature and value class (nature) of the following quadratic forms: (i) $x_1^2 + 5x_2^2 + x_3^2 + 2x_2x_3 + 6x_3x_1 + 2x_1x_2$ (ii) $x_1^2 + 4x_2^2 + x_3^2 - 4x_1x_2 - 4x_2x_3 + 2x_3x_1$ **Answer: (i) r = 3, p = 2, s = 1, Indefinite** **(ii) r = 1, p = 1, s = 1, Positive semidefinite**
C	6	Determine the value class of following quadratic forms by nature of leading principal minors. (i) $-3x^2 - 3y^2 - 3z^2 - 2xy - 2xz + 2yz$ (ii) $2x^2 + 2y^2 + 5z^2 - 2xy$ **Answer: (i) Negative definite, (ii) Positive definite**
C	7	Find the value of k for which the quadratic form $Q(x) = 5x^2 + y^2 + kz^2 + 4xy - 2xz - 2yz$ is positive definite. **Answer: k > 2**

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